

BBRES-RNG

Validation report

Bit Based Randomized Entropy System - Scheduler Based RNG

Developed by Mehul Singh | Source report date: 25 July 2026

53 / 54	31,377,615	500,000	7.999952 / 8
checks marked PASS	generated bits	integers in [0, 1000]	Shannon entropy per byte

A broad test of the implemented output

The supplied report covers bitstream behavior, integer output, distribution, structure, prediction, and entropy estimates. Most checks passed across the reported sample; the detailed results and plots follow in the full report.

Test family	Result	Test family	Result
NIST SP 800-22 core	9 / 9 PASS	Adversarial / prediction	5 / 5 PASS
NIST extended	15 / 15 PASS	Cryptographic wrapper	1 / 1 PASS
Uniformity	3 / 3 PASS	Integer sequence	8 / 8 PASS
Spectral / structure	4 / 4 PASS	Integer distribution	8 / 9 PASS

One integer-distribution result

Cramér-von Mises: $p = 0.000915$, marked FAIL ($n = 20,000$); the other eight listed integer-distribution checks pass.

Entropy estimate in context

The report gives a non-IID assessed min-entropy estimate of 0.4573 bits/bit, bound by its LRS estimator; a same-length NumPy control returned 0.4309. The source report notes estimator caveats, uses an independent SP 800-90B Section 6.3 implementation, and does not include restart testing. Treat these as reported estimates, not formal compliance or certification.

Console observation / BBRES RNG V1.2.0

Run	Elapsed	Calculated rate
20,000,000 bits	225,286.890310 s	~88.8 bits/s

Individual console runs, not a controlled benchmark. Machine/JVM conditions are not shown; the V1.2.0 build has not been verified as the same build used for the statistical sample. Rates are retained as observations, not a performance claim.

PART 1: NIST SP 800-22 Core Tests (Bits)

Test	p-value	Result	Detail
Frequency (Monobit)	0.441746	PASS	ones=15,690,962 / zeros=15,686,653 (ratio: 0.500069)
Block Frequency (M=128)	0.334105	PASS	245,137 blocks tested
Runs Test	0.842864	PASS	15,688,252 total runs
Longest Run of Ones	0.592417	PASS	M=10,000 block size
Cumulative Sums (Fwd)	0.444146	PASS	z=6,836
Cumulative Sums (Rev)	0.691439	PASS	z=5,226
Approximate Entropy (m=2)	0.831267	PASS	ApEn=0.693147
Serial (m=2) - delta1	0.729395	PASS	delta1=0.6311
Serial (m=2) - delta2	0.842781	PASS	delta2=0.0393

The NIST Special Publication 800-22 defines the gold standard battery of statistical tests for evaluating randomness quality. A p-value above 0.01 (alpha) indicates no statistically significant deviation from ideal random behavior at the 99% confidence level.

PART 2: NIST SP 800-22 Extended Tests

Test	p-value	Result	Detail
Maurer's Universal Statistical	0.940885	PASS	fn=6.1962
Poker Test (m=4)	0.332432	PASS	Chi-sq=16.7760
Random Excursions (x=-4)	0.719514	PASS	J=3,896 cycles, chi-sq=2.873
Random Excursions (x=-3)	0.942658	PASS	J=3,896 cycles, chi-sq=1.223
Random Excursions (x=-2)	0.630408	PASS	J=3,896 cycles, chi-sq=3.454
Random Excursions (x=-1)	0.114422	PASS	J=3,896 cycles, chi-sq=8.869
Random Excursions (x=+1)	0.045495	PASS	J=3,896 cycles, chi-sq=11.314
Random Excursions (x=+2)	0.805368	PASS	J=3,896 cycles, chi-sq=2.306
Random Excursions (x=+3)	0.493509	PASS	J=3,896 cycles, chi-sq=4.399
Random Excursions (x=+4)	0.524710	PASS	J=3,896 cycles, chi-sq=4.174
Random Excursion Variant	0.872715	PASS	Cycles: 3,896
Binary Matrix Rank	0.661235	PASS	Full=594, M-1=1148, rest=258 (n=2000)
Non-Overlapping Template (m=9)	0.767012	PASS	mu=612.83, blocks=100
Overlapping Template (m=9)	1.000000	PASS	lambda=612.83, blocks=100
Linear Complexity	0.298157	PASS	M=500, blocks=1000

Extended tests include Maurer's Universal Statistical test (compressibility), Poker test (block uniformity), Random Excursion Variant (cycle analysis), Binary Matrix Rank (linear dependence), Non-Overlapping and Overlapping Template Matching (pattern occurrence), and Linear Complexity (LFSR complexity).

PART 3: Distribution and Uniformity Analysis (Bits)

Test	p-value	Result	Detail
Byte-Level Chi-Square	0.370831	PASS	256 bins: min=15024, max=15667, exp=15321.1
Nibble-Level Chi-Square	0.332432	PASS	16 bins tested
2-Bit Pair Distribution	0.080433	PASS	00:3,919,285 / 01:3,921,738 / 10:3,926,344 / 11:3,921,440

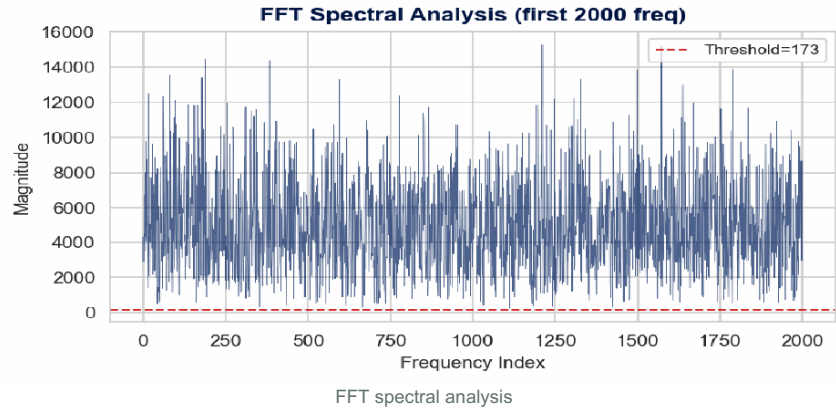
Uniformity tests verify that all possible bit patterns occur with expected frequency at 2-bit, 4-bit (nibble), and 8-bit (byte) granularities.



PART 4: Spectral and Structural Analysis (Bits)

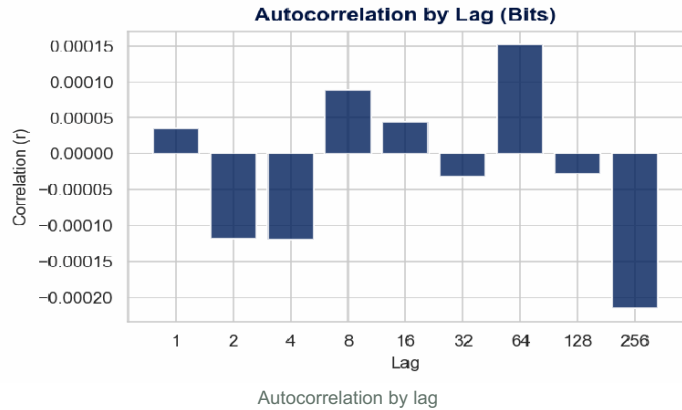
Test	p-value	Result	Detail
Spectral FFT (Periodicity)	0.959660	PASS	peaks below threshold: 14,904,398/15,688,807
Compression Ratio (zlib)	0.218383	PASS	Ratio=1.000307 (3,923,407/3,922,201)
Binary Derivative (1st order)	0.842642	PASS	ones=15,688,251/31,377,614 (ratio=0.499982)
Turning Point Test	0.887764	PASS	TPs=1307482, Expected=1307398.7

The Discrete Fourier Transform test checks for periodic components in the bitstream. peaks below threshold: 14,904,398/15,688,807. p-value: 0.959660 - No detectable periodicity.



Autocorrelation Profile

Lag	1	2	4	8	16	32	64	128	256
r	+0.0000	-0.0001	-0.0001	+0.0001	+0.0000	-0.0000	+0.0002	-0.0000	-0.0002

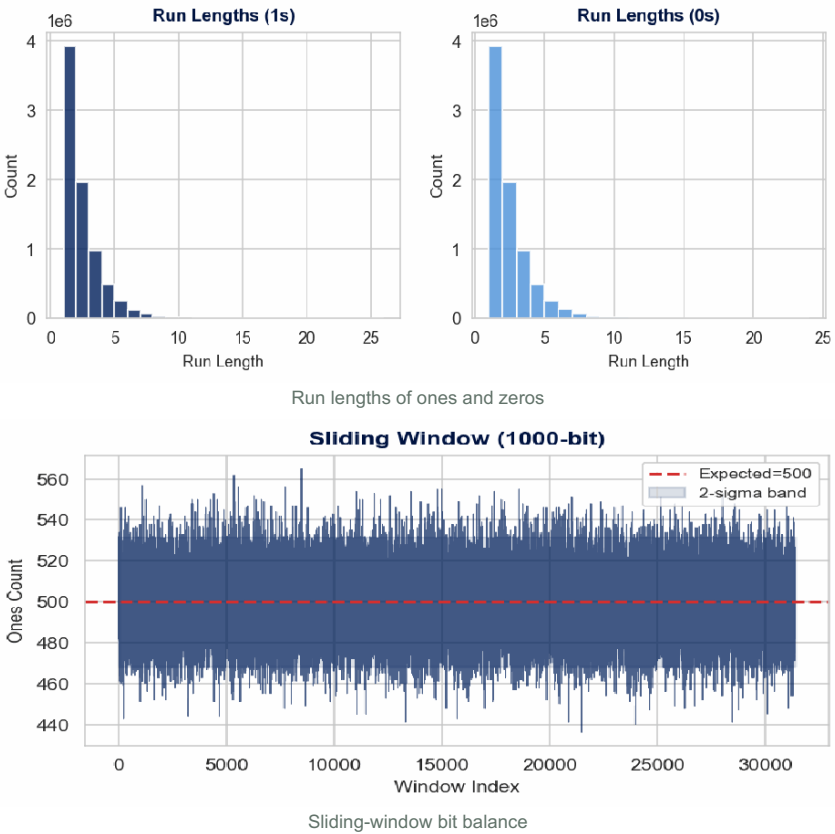


PART 5: Entropy Analysis (Bits)

Metric	Value	Theoretical Max	Assessment
Shannon Entropy (8-bit)	7.999952	8.000000	99.999% of max
Min-Entropy (8-bit)	7.967791	8.000000	99.60%
Transition Rate	0.499982	0.500000	Near-ideal

PART 6: Pattern and Run-Length Analysis (Bits)

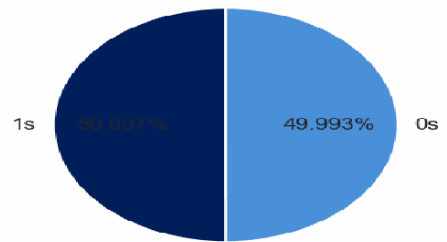
Metric	Ones Runs	Zeros Runs	Expected
Count	7,844,126	7,844,126	~7,844,403
Mean Length	2.0003	1.9998	2.0000
Max Length	25	23	~25



PART 7: Bit-Level Visual Analysis

Visual inspection provides an intuitive layer of validation. A truly random bitstream should show no visible patterns in its matrix representation, uniform density in scatter plots, and a symmetric random walk in cumulative sums.

Bit Balance (0 vs 1)



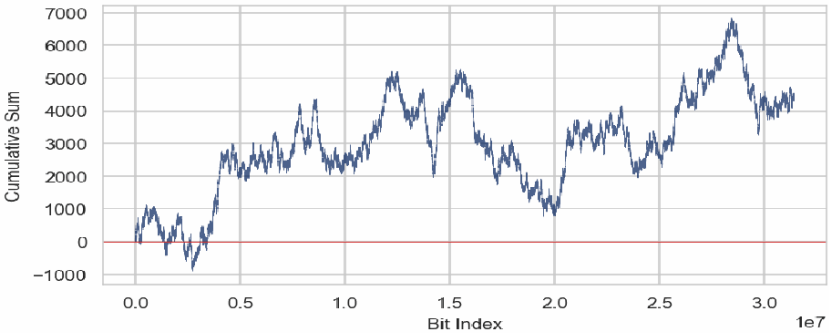
Bit balance

Bit Matrix (100x100)



100 by 100 bit matrix

Cumulative Sum Random Walk



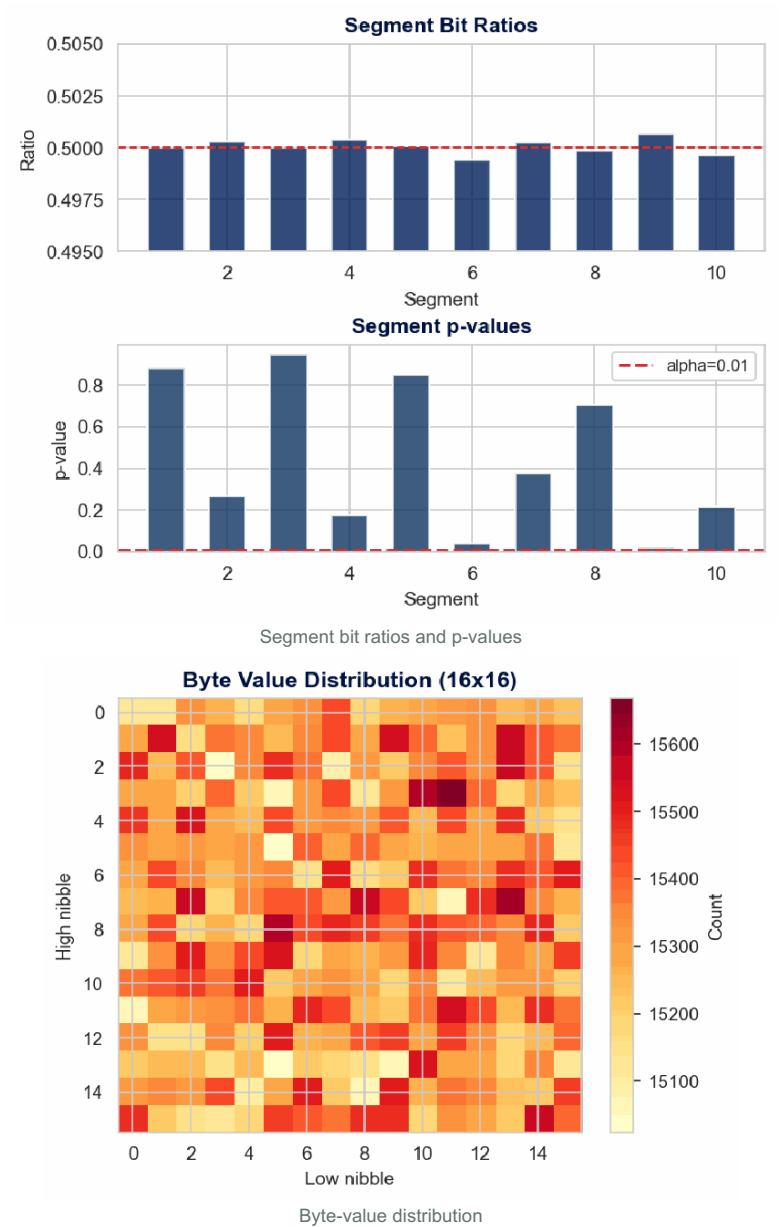
Cumulative-sum random walk

PART 8: Cross-Segment Consistency (Bits)

The bitstream was divided into 10 equal segments and each independently tested.

Seg	1	2	3	4	5	6	7	8	9	10
Ratio	0.5000	0.5003	0.5000	0.5004	0.5001	0.4994	0.5002	0.4999	0.5007	0.4997
p-val	0.881	0.267	0.947	0.173	0.849	0.040	0.376	0.704	0.019	0.217

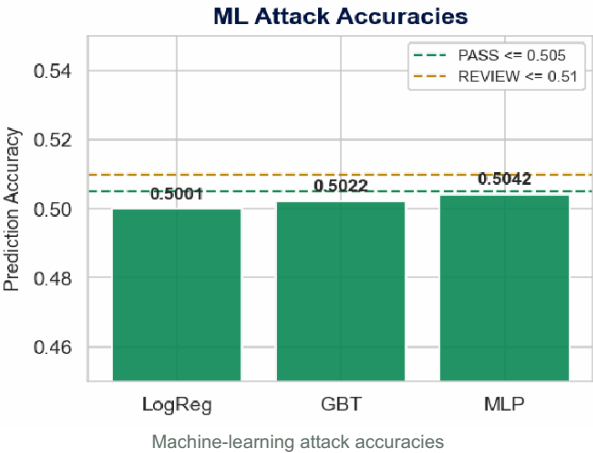
KS test on segment p-values: stat=0.2327, p=0.5741. Cross-half correlation: r = 0.000195.



PART 9: Adversarial and Predictability Tests (Bits)

Test	p-value	Result	Detail
Frequency Prediction (w=8)	0.722460	PASS	Acc: 0.5011 (expected ~0.5011) mean acc=0.5001 over 5 folds
ML Attack (LogReg, w=16)	0.486453	PASS	(range 0.4927-0.5056), n=50,000 strided samples status=PASS (PASS<=0.505, REVIEW<=0.51)
ML Attack (GBT, w=16)	0.314266	PASS	mean acc=0.5022 over 5 folds (range 0.4898-0.5129), n=40,000 strided samples status=PASS (PASS<=0.505, REVIEW<=0.51)
ML Attack (MLP, w=16)	0.001116	PASS	mean acc=0.5042 over 5 folds (range 0.5024-0.5056), n=40,000 strided samples status=PASS (PASS<=0.505, REVIEW<=0.51)
Pattern Repetition (w=63)	1.000000	PASS	No repetition detected

These tests attempt to predict the next bit using frequency-based pattern matching and machine learning (Logistic Regression, Gradient Boosted Trees, MLP Neural Network). An accuracy near 50% indicates the output is unpredictable. Pattern repetition checks for repeated subsequences.



Cryptographic Wrapper Validation

Test	p-value	Result	Detail
SHA-256 CSPRNG Wrapper	1.000000	PASS	Post-hash prediction: 0.5302

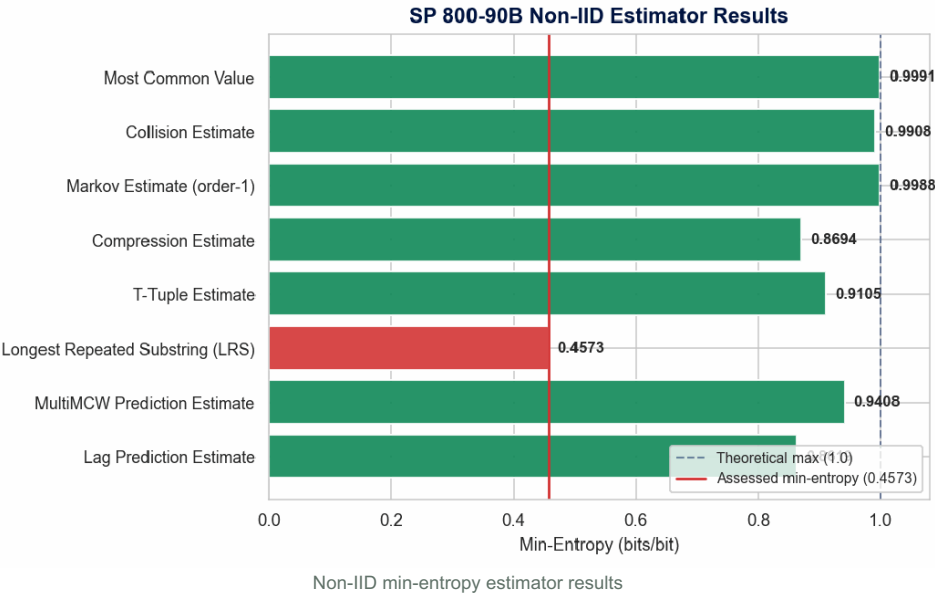
The BBRES-RNG output is seeded into a SHA-256 based CSPRNG wrapper, and the resulting secure bits are re-tested for predictability. This validates the suitability of BBRES-RNG output as entropy source for cryptographic applications.

PART 9B: NIST SP 800-90B Non-IID Min-Entropy Estimation

SP 800-22 (Parts 1-8) tests statistical distribution - whether output looks uniformly random. SP 800-90B tests a stricter, different property: min-entropy against a knowledgeable adversary - a conservative bound on how well the single best prediction strategy could do. The assessed min-entropy of a source is the minimum across all applicable estimators below, since an adversary picks whichever attack works best.

Estimator	Section	H_min (bits/bit)	Detail
Most Common Value	6.3.1	0.9991	mode_count=15,690,962/31,377,615, p_hat=0.500069
Collision Estimate	6.3.2	0.9908	adjacent-match_rate=0.500018, implied bias p_hat=0.502977 (exact reduc...
Markov Estimate (order-1)	6.3.3	0.9988	P(0)=0.499931, P(0 0)=0.499949, P(1 1)=0.500086, L=128-bit worst-path ...
Compression Estimate	6.3.4	0.8694	L=6-bit symbols, mean stat=5.2178 bits/sym, 99%-LB=5.2163 bits/sym
T-Tuple Estimate	6.3.6	0.9105	t=21, max_count=39/31,377,595
Longest Repeated Substring (LRS)	6.3.7	0.4573	longest exact repeat=48 bits (final search ceiling=62), W=49, max occu...
MultiMCW Prediction Estimate	6.3.8	0.9408	window=4095, correct=15,687,351/31,373,520 (50.0019%), longest correct...
Lag Prediction Estimate	6.3.9	0.8619	lag=89, correct=15,690,272/31,377,526 (50.0048%), longest correct run=...

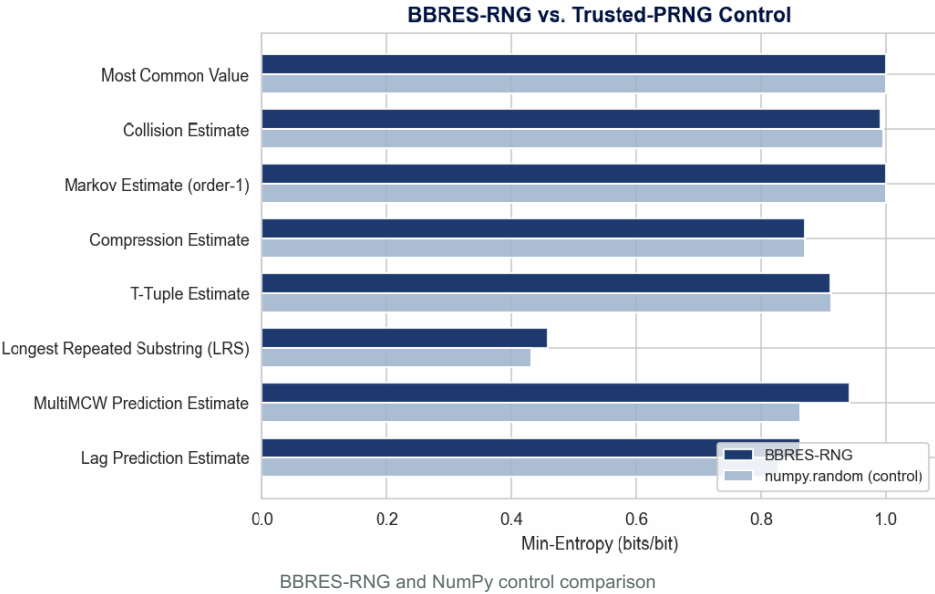
Assessed min-entropy: 0.4573 bits/bit (45.73% of theoretical max), bound by the Longest Repeated Substring (LRS) estimator.



Control Comparison: BBRES-RNG vs. numpy.random

The T-Tuple and Longest Repeated Substring (LRS) estimators are documented by NIST to produce conservative (lower) bounds on binary (2-symbol) alphabets specifically, because long tuple collisions occur by pure chance far more often with only 2 possible values than with larger symbol alphabets. To confirm a low LRS/T-Tuple score reflects this known estimator behavior rather than an actual weakness, the same estimator suite was run against a same-length sample from numpy.random (a trusted, widely-used PRNG) as a control.

Control (numpy.random) assessed min-entropy: 0.4309 bits/bit , bound by the same Longest Repeated Substring (LRS) estimator - confirming this is expected estimator behavior, not a BBRES-RNG-specific defect.



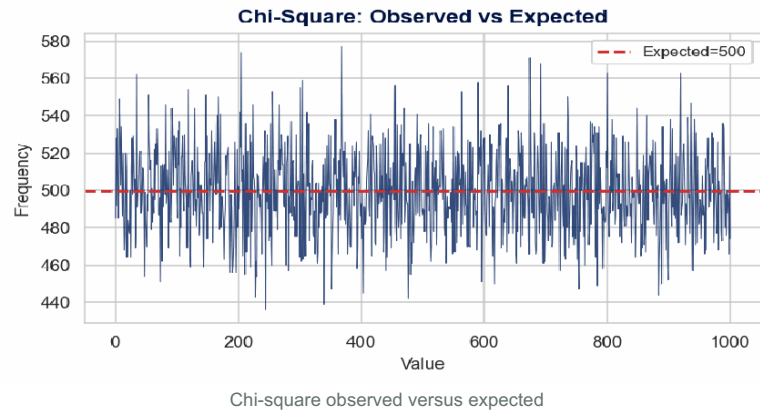
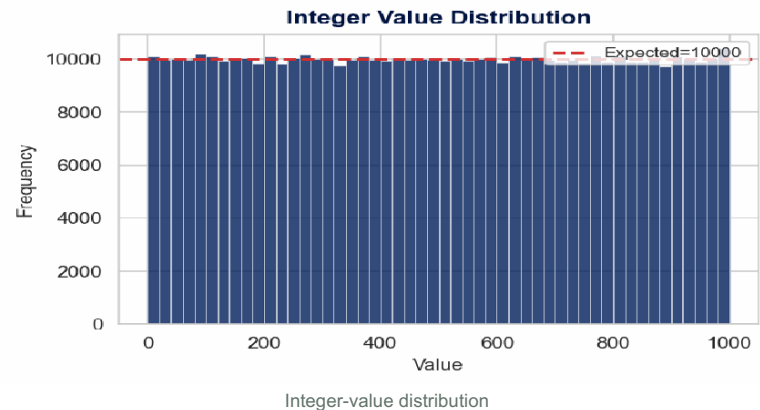
Caveats: this is an independent implementation of SP 800-90B Section 6.3 (non-IID track), not the NIST-validated reference tool, and results here should not be cited as formal 90B compliance. Notably, it does not include restart testing (Section 3.1.4) - real 90B certification requires sampling across ± 1000 independent restarts of the noise

source to catch low-entropy startup states, which a single long sequence cannot detect. For a thread-scheduling based source such as BBRES-RNG, restart testing is a natural and recommended next step.

PART 10: Integer Distribution Tests

Test	p-value	Result	Detail
Chi-Square Uniformity	0.233847	PASS	k=1001, chi-stat=1032.15
Mean (Z-test)	0.151531	PASS	Actual=499.4139, Expected=500.0000
Variance (Chi-sq)	0.775808	PASS	Actual=83547.4570, Expected=83500.0000
Binned Goodness-of-Fit	0.087485	PASS	Chi-sq=62.9084, bins=50
Birthday Spacing	1.000000	PASS	Collisions=4092, lambda=17162706.48
Coupon Collector	0.500000	PASS	Range 1001 too large
Collision Test	0.999662	PASS	Expected=15383.0, Actual=15383.0
Anderson-Darling	0.034560	PASS	A2*=0.8182
Cramér-von Mises	0.000915	FAIL	W2=0.262626, n=20,000

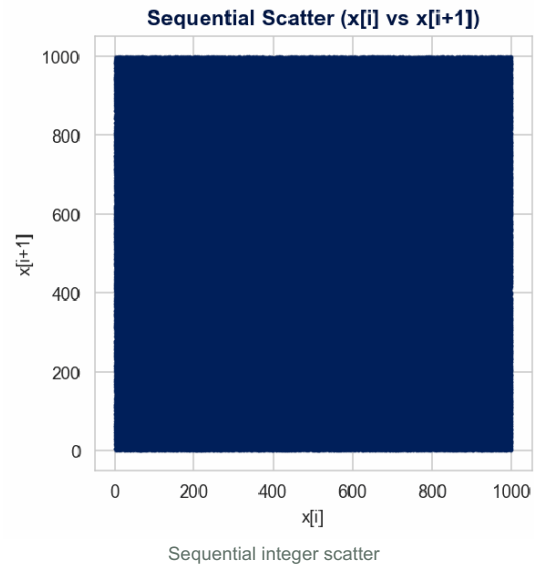
Integer output tested across range [0..1000] (1001 values). Tests include Chi-Square uniformity, binned goodness-of-fit, KS test, Birthday Spacing, Coupon Collector, Collision, Anderson-Darling, and moment analysis.

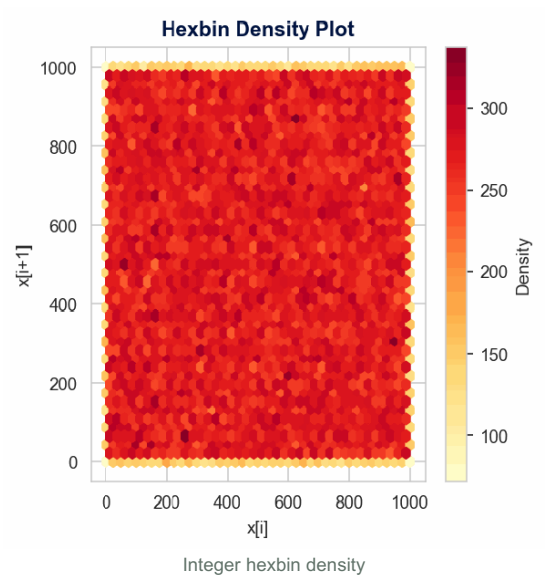


PART 11: Integer Sequence Tests

Test	p-value	Result	Detail
Skewness/Kurtosis	0.902919	PASS	Skew=0.0015, Kurt=-1.2006
Maximum-of-5	0.016055	PASS	KS=0.004910, groups=100000
Runs Up/Down	0.697900	PASS	Runs=332876, Expected=332991.7
Lag-1 Autocorrelation	0.235645	PASS	r=0.001677
Gap Test (KS)	0.918870	PASS	Mean gap=1027.03, Expected=1001
Permutation (t=5)	0.969807	PASS	120/120 patterns observed
Spearman Rank Correlation	0.235451	PASS	rho=0.001678
Median Test	0.687807	PASS	Above=249898, Below=249613, Median=499.0

Sequence tests verify that consecutive integers show no patterns, memory, or predictability. Includes runs test, multi-lag autocorrelation, gap test, permutation test, Spearman rank correlation, maximum-of-t, skewness/kurtosis, and median test.



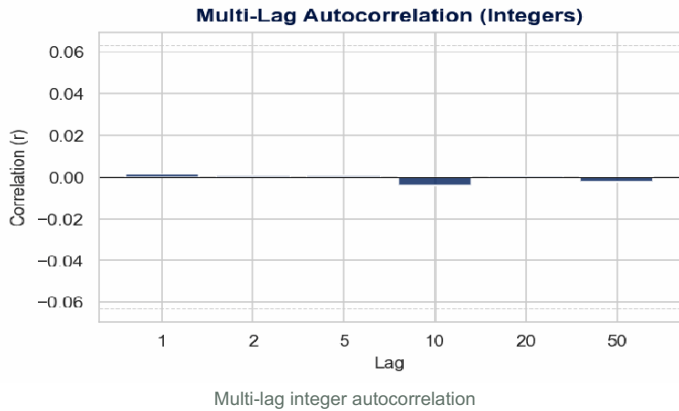


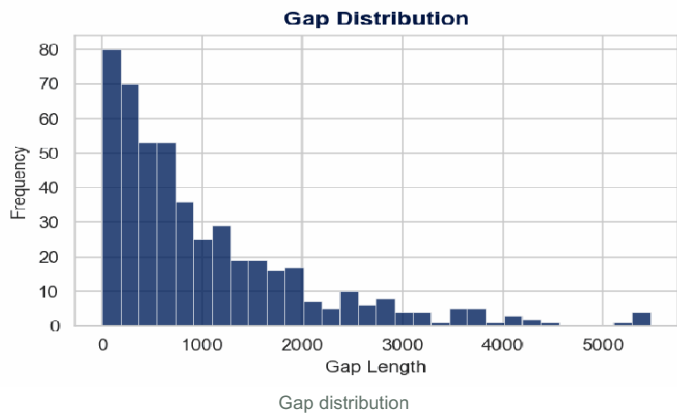
Integer Statistical Moments

Metric	Expected	Actual	Diff	p-value
Mean	500.0000	499.4139	0.5861	0.151531
Variance	83500.0000	83547.4570	47.4570	0.775808

Multi-Lag Autocorrelation (Integers)

Lag	1	2	5	10	20	50
r	+0.0017	+0.0007	+0.0006	-0.0040	+0.0003	-0.0021





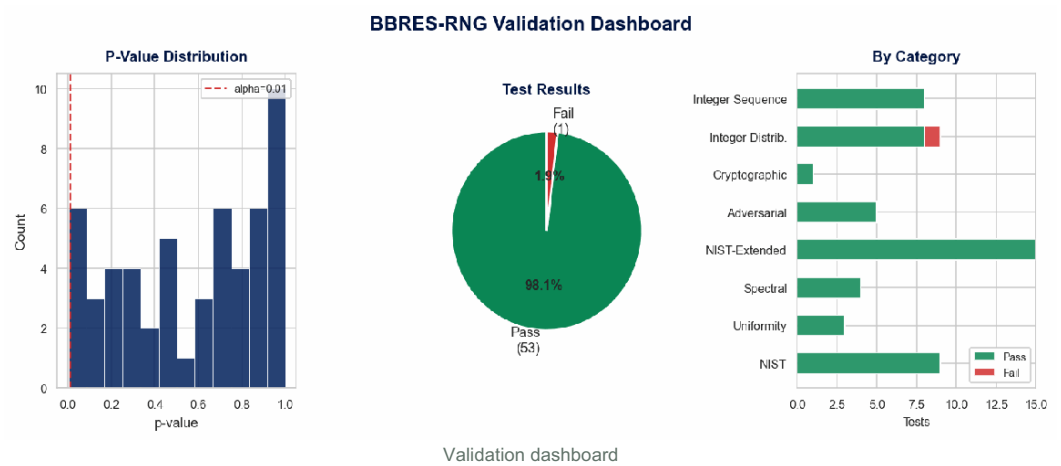
RESULTS SUMMARY

Category	Tests	Passed	Status
NIST	9	9	PASS
Uniformity	3	3	PASS
Spectral	4	4	PASS
NIST-Extended	15	15	PASS
Adversarial	5	5	PASS
Cryptographic	1	1	PASS
Integer Distribution	9	8	FAIL
Integer Sequence	8	8	PASS
GRAND TOTAL	54	53	1 FAIL

SP 800-90B assessed min-entropy: 0.4573 bits/bit (bound by Longest Repeated Substring (LRS); see Part 9B). This is a separate, adversarial-strength metric alongside the statistical PASS/FAIL battery above - see Part 9B for full context and caveats before quoting this figure on its own.

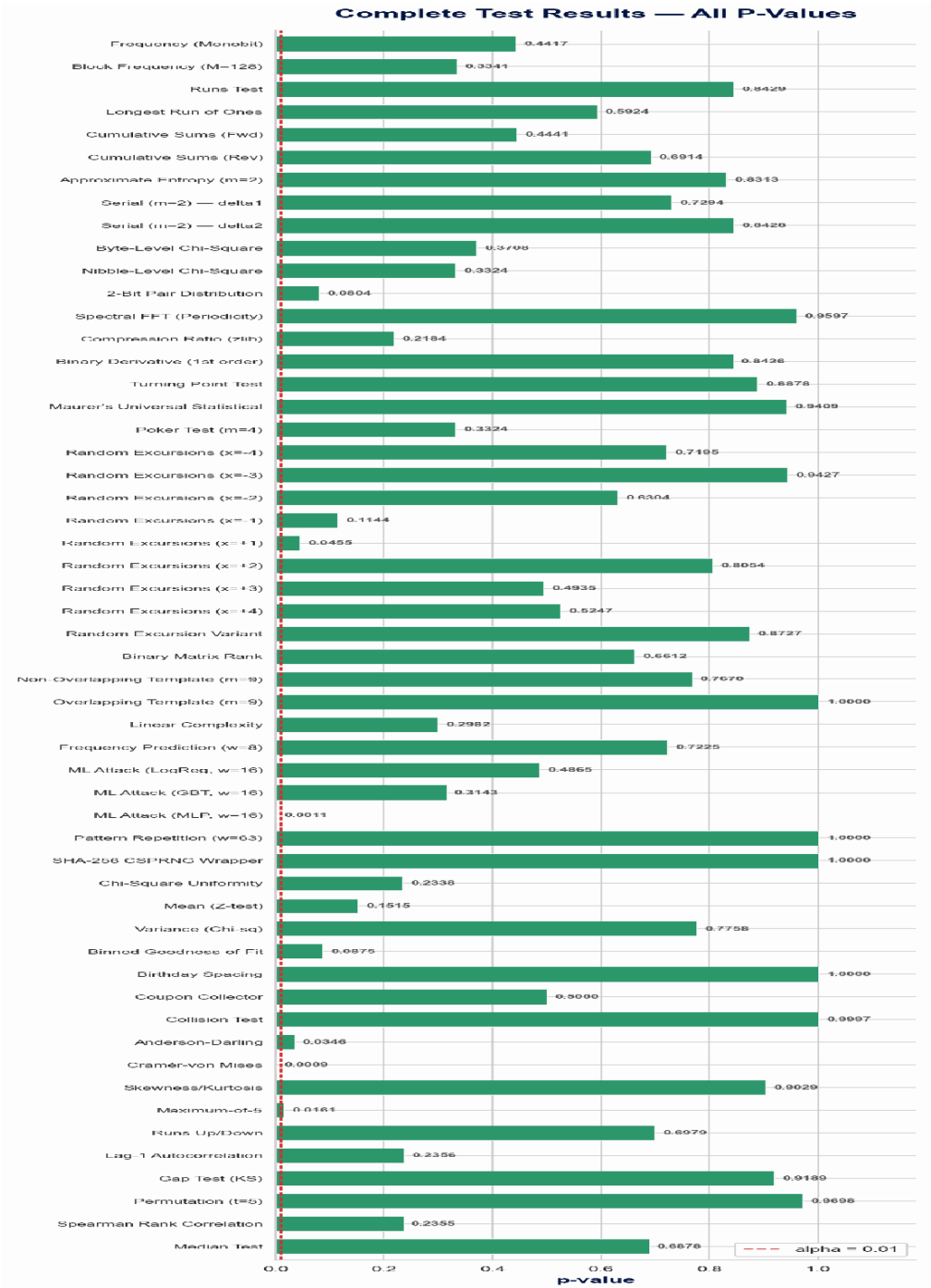
ONE EXCEPTION REPORTED

Failed tests (1): Cramér-von Mises



Complete P-Value Results

Each bar represents one test. Green bars exceed the $\alpha=0.01$ threshold (PASS). Red bars fall below (FAIL). Numeric p-values shown at bar ends.



All reported p-values